CampusIQ

Turning disconnected campus data into decisions faculty can act on

HacXLerate 2026 · Round 1

Problem statement: AI-Powered Student Analytics and Success Platform, KPMG in India

8 data sources integrated · 1,200 students scored · 2 predictive models · live dashboard

THE PROBLEM

Colleges find out too late

- A college already records everything it needs: attendance, marks, LMS activity, placement practice, clubs, feedback.
- But each sits in its own system. Nobody reads them together, so nothing adds up to a warning.
- The signal arrives as a result sheet in week 16, when nothing can be done about it.

What a mentor actually needs to hear

“Vivaan's attendance has fallen 14 points since July and he has stopped opening the LMS. He is heading for a failed course. Talk to him this week.”

Today that sentence exists nowhere, because no single system can form it.

Seven systems, none of them talking

Academic	CGPA, marks, backlogs
Attendance	subject-wise registers
LMS	logins, submissions
Engagement	events, clubs, certifications
Placement	aptitude, coding, mocks
Skills	technical, soft skills
Feedback	student and faculty

OUR SOLUTION

One record per student, and a reason to act

1 Integrate

8 source files, cleaned and joined on a single student ID

2 Score

Student Success Score 0-100 from 7 weighted pillars

3 Flag

Rules catch what is wrong today; two models predict who slips next

4 Act

Every flag carries its reasons and a suggested intervention

What the brief asked for, and where it lives

✓ Data integration across 7 categories	Data Integration page, with a full cleaning log
✓ Student Success Score	7 pillars, published formula, shown on every page
✓ At risk: academic and placement	Two separate models, one per outcome
✓ Interactive dashboard	8 views, filterable by department and year
✓ Bonus: segmentation + explainable score	7 segments; per-student score drivers

DATA INTEGRATION · 30%

Real exports are messy, so ours are too

- Eight separate source files, deliberately dirty: IDs written four different ways, duplicate rows, CGPA typed as a percentage, “AB” in a marks column, attendance above classes held, students missing from whole systems.
- The pipeline fixes each one and logs it, so nothing is changed silently.

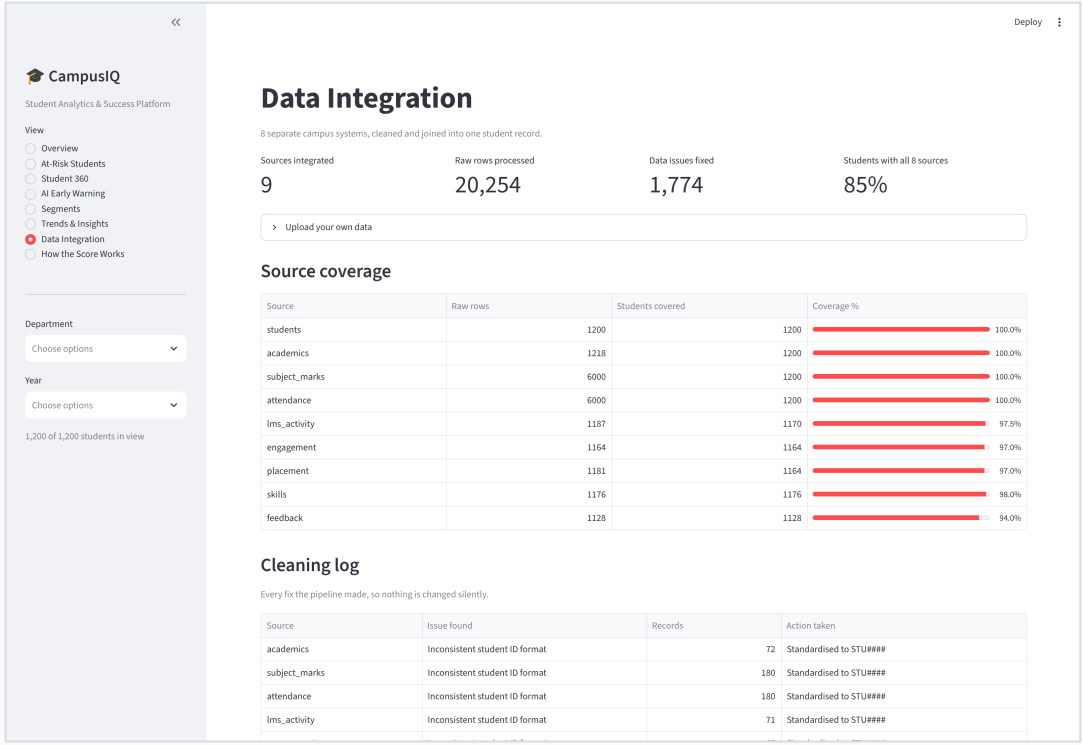
8
sources integrated

13,000+
raw rows processed

900+
data issues fixed and logged

Insight only the combined data shows:

Students low on attendance AND assignments average 48 marks; students high on both average 73. Neither register shows that gap alone.



SUCCESS SCORE · 25%

A score faculty can check by hand

Seven pillars, each scored 0-100 from raw indicators, combined with fixed weights.



Risk flags use the score AND six triggers

so a decent overall score cannot hide one serious problem

- Attendance below 65%
- 2 or more backlogs
- CGPA below 5.5
- Under half of assignments submitted
- No LMS login for 14+ days
- Attendance down 10+ points in 3 months

Explainable by construction

Because the score is a weighted sum, every student's total breaks down into points per pillar. The dashboard shows which pillars pull each student above or below the campus average.

Two models, because there are two risks

- Trained on last year's cohort of 1,500 students, whose year-end outcomes are already known.
- The brief names two risks, and they are different groups: a student can be heading for a good degree and still be unprepared for placement.

Academic risk model

Predicts: failed course, dropout or formal intervention

78% of cases caught

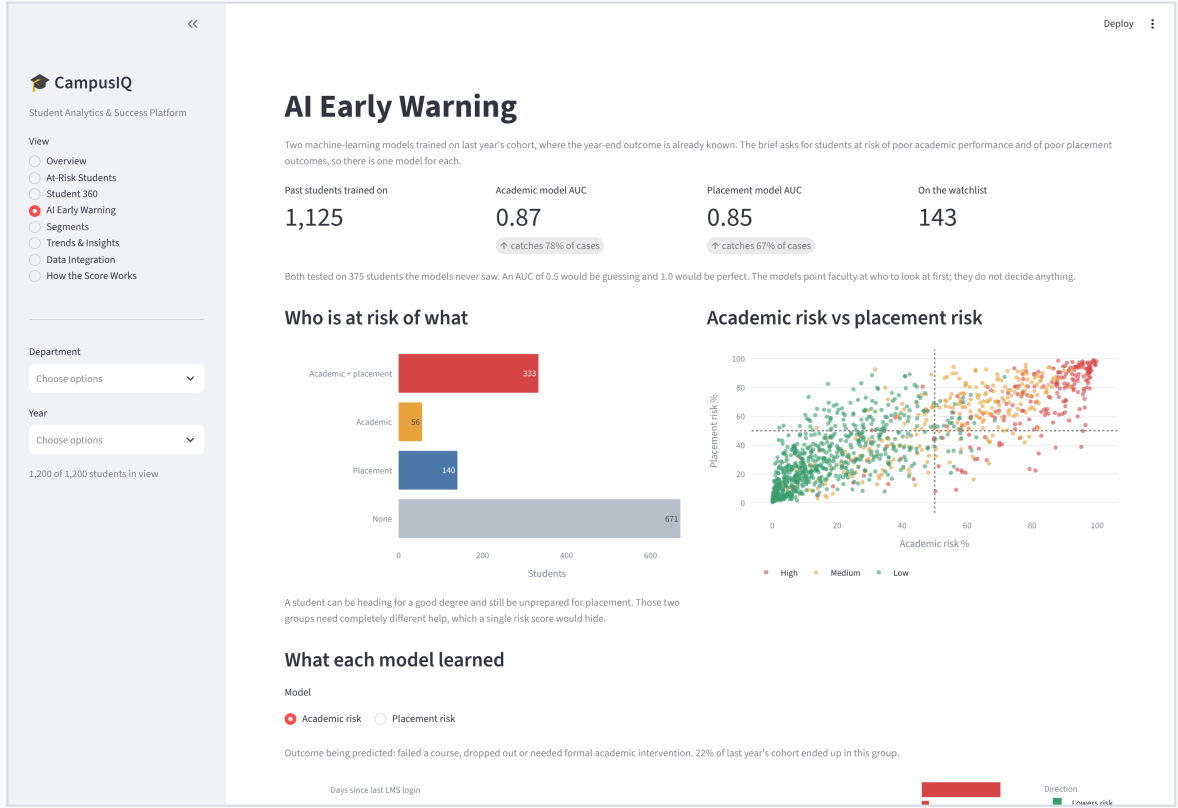
0.87
AUC, unseen students

Placement risk model

Predicts: no offer or not placement-ready

67% of cases caught

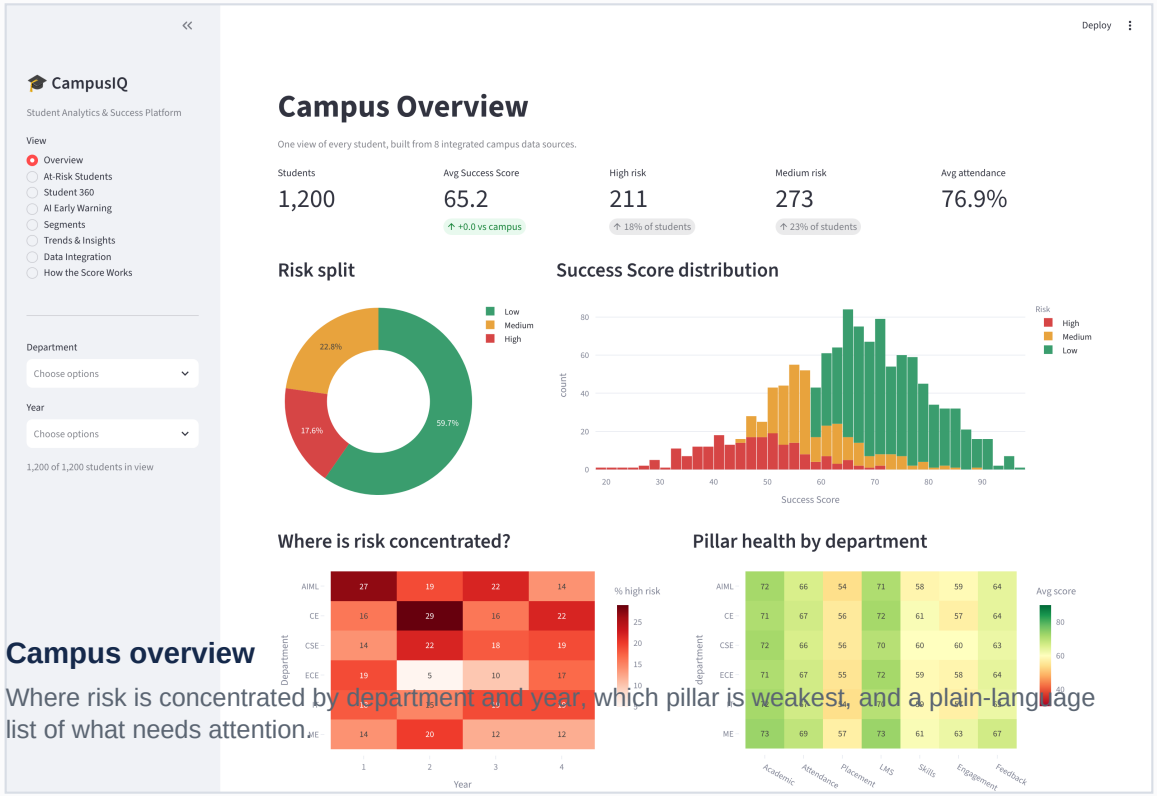
0.85
AUC, unseen students



Logistic regression on purpose, not a black box: the coefficients are shown to faculty as the reason behind every probability, which a college has to be able to defend.

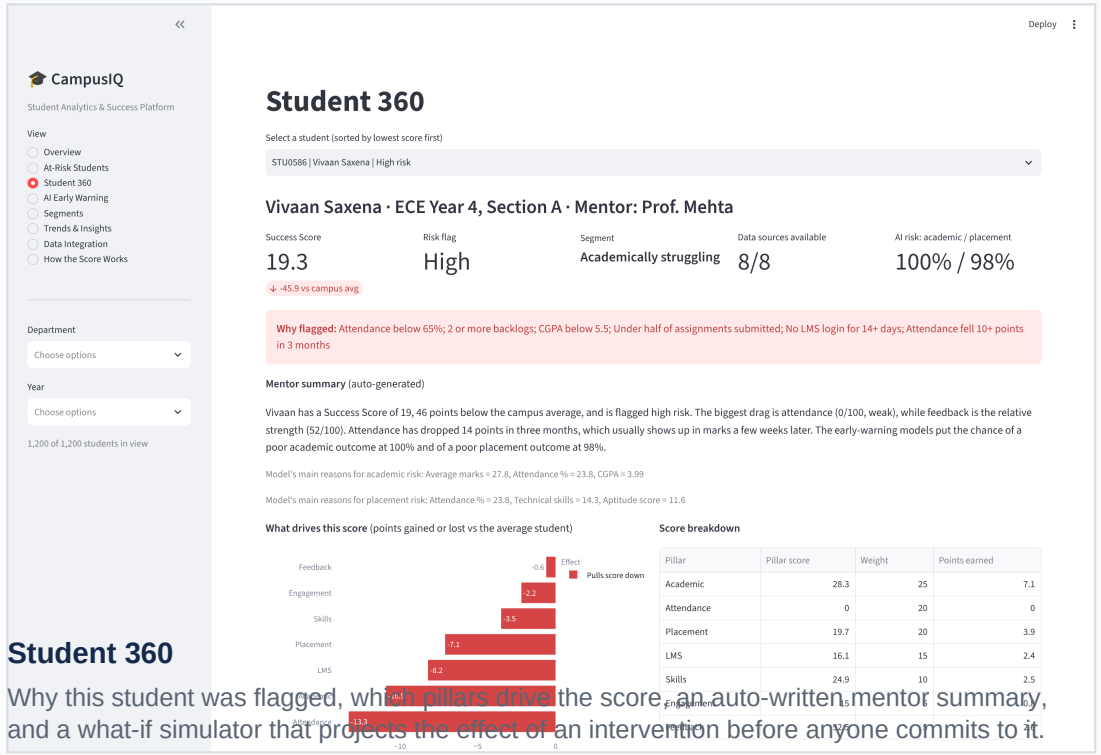
DASHBOARD · 25%

Built for a mentor with ten minutes



Campus overview

Where risk is concentrated by department and year, which pillar is weakest, and a plain-language list of what needs attention.



Student 360

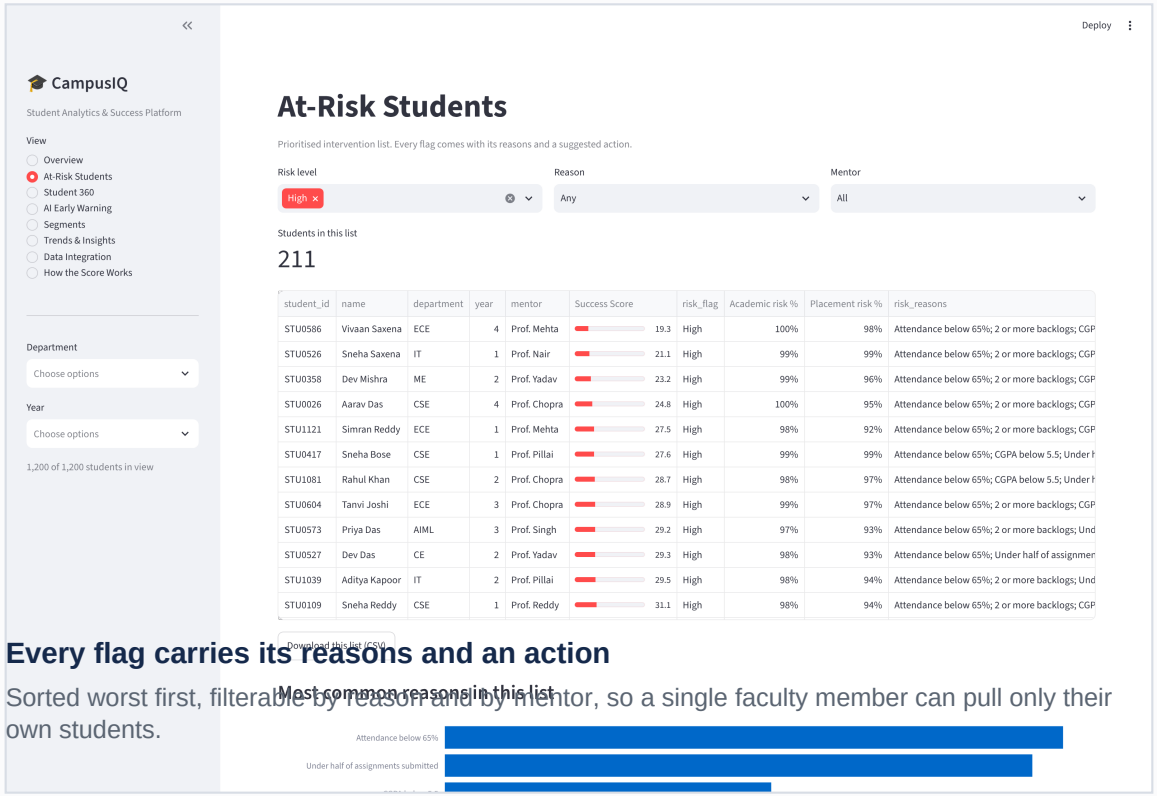
Why this student was flagged, which pillars drive the score, an auto-written mentor summary, and a what-if simulator that projects the effect of an intervention before anyone commits to it.

8 views · Overview · At-Risk Students · Student 360 · AI Early Warning · Segments · Trends & Insights · Data Integration · How the Score Works

Every list exports to CSV, because the next step happens in a mentor meeting, not in a browser tab.

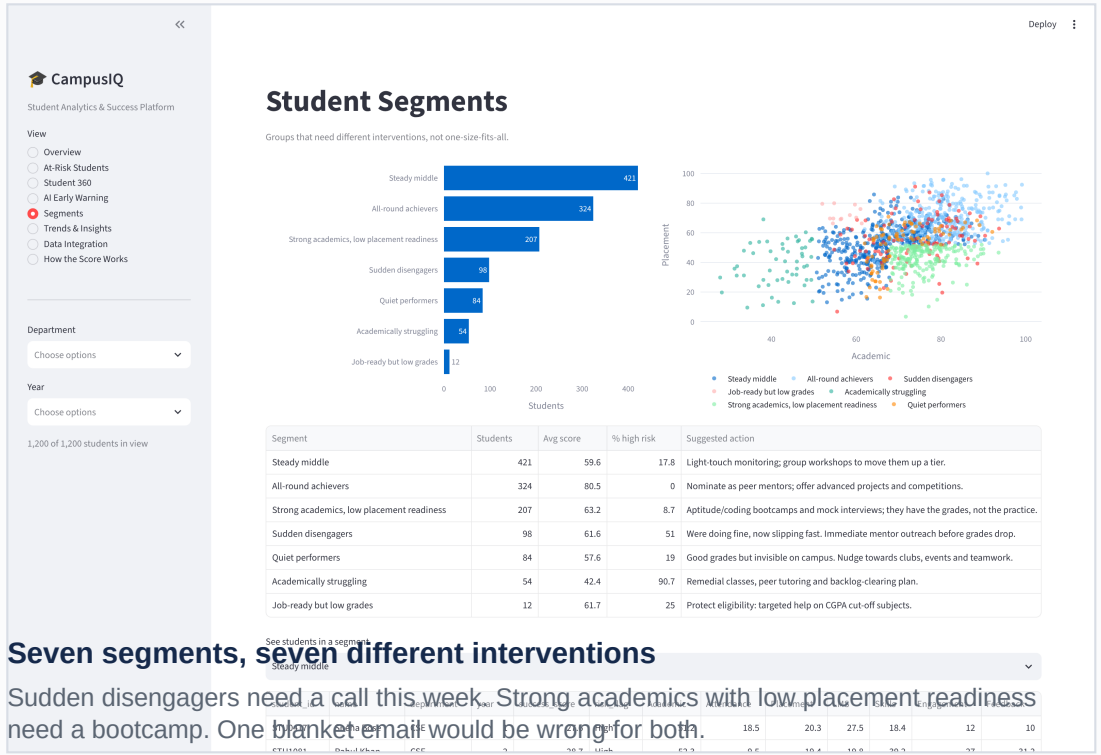
DECISION INTELLIGENCE, NOT A DASHBOARD

From a flag to a next step



Every flag carries its reasons and an action

Sorted worst first, filterable by reason and by mentor, so a single faculty member can pull only their own students.



Seven segments, seven different interventions

Sudden disengagers need a call this week. Strong academics with low placement readiness need a bootcamp. One blanket email would be wrong for both.

The sharpest result

143 students are Low risk by every rule, yet a model puts them above 50%. A threshold report would miss all of them. That is the whole point of adding the models.

IMPACT

What it changes, and what comes next

For a college, today

- Week 3 instead of week 16: the warning arrives while a semester can still be changed.
- A mentor opens one page instead of chasing five systems, and leaves with a named action.
- Scarce help goes where it works: the 56 academic-only cases and the 140 placement-only cases get different support.
- Every number is checkable, so faculty can disagree with a flag and see exactly why it fired.

What we would build next

- Live connectors to the college ERP and LMS, replacing the CSV upload we built for Round 1.
- Retrain each semester as real outcomes arrive, and report accuracy openly.
- Mentor workflow: assign an intervention, record what happened, measure whether it worked.
- Fairness checks by department, gender and entry background before any college acts on a flag.

Honest limitation

All data here is synthetic: no college can hand out real student records for a hackathon. The pipeline, scoring and models are built to run on real exports unchanged, and the dashboard already accepts an uploaded CSV in place of any source file.

Try it yourself

No login needed. Open Student 360 and pick the first student in the list.

Live dashboard	see submission for the link
Source code	see submission for the GitHub link
Success Score note	docs/success_score_note.md in the repository